

Infrastructure Provisioning Checklist

Multi-Agent Orchestrator — AI Banking Assistant
AWS Account: 861976376325 | Region: us-east-1

1. Prerequisites

- AWS CLI installed and configured with valid credentials
- Python 3.9+ installed with boto3 and botocore packages
- IAM permissions: CloudFormation, DynamoDB, IAM, Bedrock (read)
- Access to AWS account 861976376325 in us-east-1
- Git repository cloned with infra/ directory present
- Environment variables or AWS profile configured

2. CloudFormation Stack Deployment

2.1 Template Validation

- Template file exists at **infra/cloudformation.yaml**
- Template passes **aws cloudformation validate-template**
- AccountsTable resource defined (AWS::DynamoDB::Table)
- Partition key: account_id (String/S), HASH key
- BillingMode: PAY_PER_REQUEST
- OrchestratorPolicy resource defined (AWS::IAM::ManagedPolicy)
- Policy name: MultiAgentOrchestratorPolicy
- CAPABILITY_NAMED_IAM acknowledged

2.2 Stack Tags

- Tag **Project** = AIBankingAssistant on all resources
- Tag **Feature** = MultiAgentOrchestrator on all resources

2.3 Stack Outputs

- Output: AccountsTableArn (DynamoDB table ARN)
- Output: AccountsTableName (DynamoDB table name)
- Output: OrchestratorPolicyArn (IAM policy ARN)

2.4 Deployment Status

- Stack status: CREATE_COMPLETE or UPDATE_COMPLETE
- No stack drift detected
- Idempotent: re-running returns SKIPPED (no errors)

3. IAM Policy Configuration

3.1 Policy Structure

- Policy name: **MultiAgentOrchestratorPolicy**

- Policy contains exactly **2 statements**
- Statement 1 — Sid: AccountsTableReadAccess
 - Action: dynamodb:GetItem (only)
 - Resource: arn:aws:dynamodb:us-east-1:861976376325:table/Accounts
- Statement 2 — Sid: BedrockIntentClassification
 - Action: bedrock:InvokeModel (only)
 - Resource: arn:aws:bedrock:us-east-1::foundation-model/anthropic.claude-haiku-4-5-20251001-v1:0

3.2 Policy Behavior

- Duplicate creation returns SKIPPED (EntityAlreadyExists handled)
- Optional attach_to_role works when role name provided
- Follows principle of least privilege

4. Bedrock Model Access Verification

- Model ID: **anthropic.claude-haiku-4-5-20251001-v1:0**
- Region: us-east-1
- get_foundation_model returns successfully (not bedrock-runtime)
- Model lifecycle status: **ACTIVE**
- Model ARN returned in verification result
- Verification is read-only (no resource modifications)
- Handles ResourceNotFoundException gracefully
- Handles AccessDeniedException with descriptive message

Note: Model access must be enabled manually via the AWS Bedrock console.

5. DynamoDB Table & Test Data

5.1 Table Status

- Table name: **Accounts**
- Table status: ACTIVE
- Partition key: account_id (String)
- Billing mode: PAY_PER_REQUEST (on-demand)

5.2 Seed Data Verification

Account ID	Balance	Currency	Type	Status
ACC-1001	\$5,250.75	USD	savings	■ Verified
ACC-1002	\$1,200.00	USD	checking	■ Verified
ACC-1003	\$48,000.00	USD	investment	■ Verified

- Balances stored as Decimal (not float) in DynamoDB
- overwrite=False skips existing records
- overwrite=True replaces existing records
- batch_writer used for efficient writes

6. Resource Verification (verify_all_resources)

- DynamoDB table status check: returns ACTIVE
- Seed data count check: scan returns ≥ 3 items
- Bedrock model access check: model is ACTIVE
- IAM policy existence check: get_policy succeeds
- Each check is independent (wrapped in try/except)
- All checks are read-only (no modifications)
- Returns list of ProvisioningResult objects

7. CLI Script (setup_infra.py)

- **--action deploy**: Full workflow (CFN → Bedrock → Seed → Verify)
- **--action verify**: Read-only verification only
- **--action seed**: Seed test data only
- **--region**: Defaults to us-east-1, overridable
- Summary report printed with colored pass/fail output
- Exit code 0 on successful execution
- Idempotent: multiple runs produce same end state

8. Configuration Defaults (InfraConfig)

Parameter	Default Value
region	us-east-1
account_id	861976376325
table_name	Accounts
stack_name	MultiAgentOrchestratorInfra
bedrock_model_id	anthropic.claude-haiku-4-5-20251001-v1:0
policy_name	MultiAgentOrchestratorPolicy

- All defaults match production values
- All fields overridable via constructor

9. Error Handling & Idempotency

- CloudFormation: FAILED status returned on stack errors (stack left intact)
- IAM: EntityAlreadyExists → SKIPPED (no duplicates)
- Bedrock: Model not found → FAILED with descriptive message
- DynamoDB: Write failures return FAILED with partial count
- Verification: Each check independent (one failure $\neq$ all fail)
- Network/credential errors: Propagated as FAILED with exception message
- Full provisioning is idempotent (safe to re-run)

10. Testing & Validation

- Unit tests pass: InfraConfig, ProvisioningResult, ProvisioningStatus
- Unit tests pass: _stack_exists, deploy_cloudformation_stack
- Integration test: setup_infra.py --action verify exits 0
- Integration test: DynamoDB table ACTIVE, seed data present
- Integration test: Bedrock model access ACTIVE
- Integration test: IAM policy exists
- End-to-end: ACC-1001 balance = \$5,250.75 after seed
- Idempotency: Second run → all COMPLETED or SKIPPED

Sign-off

Role	Name	Date	Signature
Developer			
DevOps Lead			
Security Reviewer			